

1. **Scenario:** A program processes a list of numbers and needs to find the largest value.
Write logic to identify and return the largest number from a given list.

Answer:

- Store the numbers in a list.
- Assume the first number (0) is the largest.
- Check each number in the list one by one.
- If a number is greater than the current largest number, update the largest number.
- Continue until all numbers are checked.
- Return or print the largest number.

2. **Scenario:** A company provides employees with a 10% bonus if their salary exceeds \$50,000. Write logic to determine the bonus amount based on the given salary.

Answer:

- Ask the employee salary
- If it is exceed given amount - add 10% bonus
- Else - dont add it
- Print result

3. **Scenario:** A program evaluates a number to determine if it is even or odd.
Write logic to check whether a given number is even or odd.

Answer:

- Ask number to enter
- If $\text{num} \% == 0$ - number is even
- Else - odd
- Print the result

4. **Scenario:** A text-processing tool reverses a given word or sentence for formatting purposes.
Write logic to take a word or sentence as input and produce its reversed version.

Answer:

- Convert the input into a list of characters.
- Reverse the order of characters.
- Join the reversed characters into a string.
- Return the reversed word or sentence.

5. Scenario: A grading system determines whether a student has passed or failed based on their score.

Write logic to check if a student has passed a subject by scoring at least 40 marks.

Answer:

- Get the subject marks
- If the student marks ≥ 40 then print it as Pass
- Otherwise, Fail

6. Scenario: A retail store offers a 20% discount if a customer's total order exceeds \$100.

Write logic to calculate the final amount to be paid after applying the discount.

Answer:

If the customer purchased - $\geq$ \$100 then make it as 20% discount

Otherwise no discount

7. Scenario: A banking system processes withdrawal requests and ensures the user has enough balance.

Write logic to check if a user has enough balance before allowing a withdrawal and update the remaining balance accordingly.

Answer:

- Read the account balance and withdrawal amount.
- If the withdrawal amount is less than or equal to the balance, process the withdrawal.
- Subtract the withdrawal amount from the balance and return the updated balance.
- If the withdrawal amount exceeds the balance, print "Insufficient funds."

